Multi-Stage Pharmaceutical Intelligence Initiative

Class-Wide GLP-1 Ocular Safety Audit & Deep Learning Screening Architecture

STAGE I TO STAGE V COMPREHENSIVE PRESENTATION

Methodological Architecture Lifecycle

Stage I & II: Discovery

Programmatic FAERS API ingestion (2023–2025).
Symptom Vectorization via SBERT.
Unsupervised Manifold Learning (UMAP/HDBSCAN).

Stage III: Validation

Disproportionality Testing (ROR, PRR, IC025).
Cohort Stratification & Demographic Demolition.
Isolation of Causal Mechanisms.

Stage IV & V: Solution

Deep Learning ONH Segmentation Framework.
Multi-modal Feature Engineering Dashboard.
Enterprise Scalability Blueprint.

Stage II: Discovery of the Acute OIN Complex

The Unsupervised Core Fraction

Processing 170,000+ unstructured safety payloads using **SBERT Encodings** and density projection isolated a latent statistical signal completely distinct from background diabetic retinopathy rules.

- **Cluster 23 + 132 Integration:** Combined into a highly purified Ocular Ischemic Syndrome cohort.
- **Sequence Rule Cleansing:** Explicitly eliminated confounding background disease references.

+193%

Phenotype Reporting CAGR (2023-2025)

Breached the mathematical noise threshold wall in late 2024

Stage III: Semaglutide Risk Matrix Stratification

Stratification Layer	ROR Metric	IC025 Bound	Signal Threshold Status
High-Dose Subcutaneous (≥ 1.0 mg)	1.751	0.421	🚨 CRITICAL SIGNAL
Low-Dose Subcutaneous (< 1.0 mg)	0.590	-0.812	⚪ Neutral Baseline
Diabesity Comorbid Profile	3.624	1.104	🚨 EXPONENTIAL SYNERGY
Pure Obesity Indication	2.392	0.766	🚨 INDEPENDENT THREAT

The Chronic Exposure Maintenance Wall

Time-to-Event Latency Partitioning

Disproportionality analysis indicates that ocular ischemic risk requires sustained cumulative baseline exposure, entirely absent during early titration phases.

1–90 Days

Titration Onboarding

No Signal Detected

91–180 Days

ROR = 0.85

Statistically Insignificant

181+ Days

ROR = 4.83

🚨 Maintenance Peak

Demographic Signals & The Cholesterol Paradox

Pre-Elderly Epicenter

ROR = 2.154 | IC025 = 0.598

Ages 51–64 represent the peak risk segment due to aggressive dose titrations compared to older brackets.

Cholesterol Paradox

No-Lipid ROR = 2.615 ($p \approx 10^{-20}$)

Risk is higher in patients with completely clean cardiovascular baselines, proving independent drug-induced vascular shifts.

Ocular Negative Control

Surface Ocular ROR = 0.98

Unbound superficial pathologies collapse back to baseline neutrality, completely invalidating general over-reporting bias.

Geriatric Prioritization: Organ-System Breakdown

Vascular vs Mechanical Safety Study (65+)

Head-to-head comparison within the high-dose weight loss stratum shatters traditional clinical tracking assumptions.

The tissue-specific perfusion crisis happens silently within the eye compartment before precipitating any generalized physical falls.

Phenotype Target	ROR	IC025 Bound
Cluster 11 (Ocular Stroke)	3.975	0.469 (🚨 YES)
Cluster 16 (Mechanical Falls)	1.555	-1.165 (⚪ NO)

Stage III: Tirzepatide Class-Wide Validation

Q1 2025 Breakout Inflection

Programmable tracking captured a rapid velocity breakout where posterior Ocular Stroke separated cleanly away from traditional Retinal Damage paths.

The 10.0 mg Titration Barrier

Dose-response non-linearity reveals an explosive risk spike specifically at the 10.0 mg saturation gateway ($p=0.004$).

The Switcher Smoking Gun

~20.0 PRR

Astronomical risk magnification in patients actively transitioning between different GLP-1 medications due to **Vascular Shock Kinetics**.

Stage IV: Neural Network Screening Architecture

U-Net Framework + ResNet-34 Encoder

To proactively shield at-risk clinical cohorts, an automated computer vision screening engine was engineered to target the underlying anatomical risk factor: **The Crowded Disc**.

- **Optic Disc Validation:** 0.9949 AUC Area
- **Optic Cup Delineation:** 0.9414 AUC Area
- **Recall Sensitivity Optimization:** ~0.97 target tracking achieved via forced dual-thresholding maps.

Automated Morphometric post-processing

Biomarker Asset	Excavated Core (≥0.2)	Congenital Cupless (0.0)
Median Tortuosity	1.53 baseline	**1.64 (High Stress)**
Peripapillary Density	53.90%	52.50% (Uniform)

*Uniform capillary density rules out pre-existing starvation; proving OIN is a pure **acute mechanical perfusion pressure collapse** inside tight anatomical channels.

Stage V: Streamlit Frontend Translation MVP

Image-Derived Clinical Decision Support

The interactive front-end application removes complex tensor logic from the clinician's workspace, providing immediate risk stratification:

Intake Context Controls: Toggle brand specifications, age criteria, and crucial binary **Switcher Status** indicators.

Interactive Overlay Canvas: PIL/Matplotlib wrapper layer displaying original fundus snapshots next to checkbox segmentation masks.

Dynamic Advisory System: State-driven logic auto-generating mandatory clinical alert boxes based on calculated metrics.

FUTURE MLOPS INTEGRATION BLUEPRINT

- **Inference Containerization:** PyTorch weights wrapped in FastAPI microservice layer inside Docker containers.
- **Drift Tracking Pipeline:** Evidently AI mapping feature activations against Prometheus/Grafana alert hooks.
- **Continuous Learning:** 80% locked historical ground truth blended with 20% manual audit updates to block catastrophic forgetting signatures.

Strategic Decisions & Clinical Mandates

1. Pre-Screening Gates

Mandatory automated fundus screening implemented prior to prescription approvals for all concurrent **Diabesity** or **inter-class GLP-1 Switcher** candidates.

2. The 10.0 mg Deferral Trigger

The computer vision app serves as a regulatory gatekeeper before step-up dose titration. If calculated baseline tortuosity exceeds 1.60, titration must be clinically deferred.

3. Objective AI Superiority

Geriatric evidence completely invalidates patient-reported frailty/dizziness tracking. Objective structural mapping is the **only valid defense** against silent microvascular collapse.

Capstone Initiative Defense Ready

"This framework successfully bridges large-scale unsupervised pharmacovigilance discovery with targeted convolutional neural network interventions—proving how predictive AI can safeguard patient vision across the modern metabolic therapeutic landscape."

Vatsal Jani | Automation & Data Science Specialist